

Quadratic Equations (Algebra)

Rajat Kalia

16 August, 2026

Infrastructure

#764, 41-A, Chandigarh {TC(official)} - 160036
 Shamlat (Ramgarh Sardaran), Ludhiana, Punjab
 SCO 24(TF), 41-D, Chandigarh - 160036
<https://www.alphaclasses.com/>

Educational

<https://github.com/rajatkalia1982>
<https://sourceforge.net/u/rajatkalia/profile>
<https://www.youtube.com/@rajatkaliaeducation>

Store

<https://www.lulu.com/spotlight/rajatkalia11>
<https://amazon.com/author/rajatkalia>

Sports

<https://www.youtube.com/@indiangamersorganization>
<https://www.facebook.com/groups/igohn>
<https://india.voobly.com/>

Social

<https://www.goodreads.com/sardarbhagatsingh>
<https://x.com/RajatKalia1>
<https://www.linkedin.com/in/rajatkalia007/>
<https://www.facebook.com/rajatkalia007>
<https://www.instagram.com/rajatkalia007/>
<https://www.twitch.tv/rajatkalia>
<https://www.facebook.com/rajatkaliaeducation>
<https://www.researchgate.net/profile/Rajat-Kalia>
<https://www.w3profile.com/rajatkalia/>
<https://www.reddit.com/user/rajatkalia/>

Blogs

<https://rk-e.blogspot.in/> [Physics]
<https://chem-rke.blogspot.com/> [Chemistry]
<https://biosciences-rke.blogspot.com/> [Bio-Sciences]
<https://rk-edu.blogspot.com/> [Mathematics]
<https://medlife-rke.blogspot.com/> [Medical Sciences]
<https://computers-mke.blogspot.com/> [Computers Engineering]
<https://ee-mke.blogspot.com/> [Electrical Engineering]
<https://me-mke.blogspot.com/> [Mechanical Engineering]
<https://ce-mke.blogspot.com/> [Civil Engineering]
<https://mke-whj.blogspot.com/> [White Hat Junior]
<https://ba-xpn.blogspot.com/> [Bachelor of Arts]
<https://constitution1982.blogspot.com/> [Constitution]
<https://debk-mke.blogspot.com/> [Accounting]
<https://comm-pke.blogspot.com/> [Commerce]
<https://brigade-13.blogspot.com/> [Air Force, India]
<https://battalion-7.blogspot.com/> [Armed Forces]
<https://design-mke.blogspot.com/> [Design]
<https://tmm18.blogspot.com/> [Meditation]

Freelancing

<https://www.guru.com/freelancers/rajat-kalia>
<https://www.codecademy.com/profiles/rajatkalia>
<https://leetcode.com/u/rajatkalia1/>
<https://www.scribd.com/user/308680256/Rajat-Kalia>

Contents

I	Theory	3
II	Problem Set	4
1	JEE Mains Problems	5

Part I

Theory

Part II

Problem Set

Chapter 1

JEE Mains Problems

Q1: Let $\alpha, \alpha + 2, \alpha \in \mathbb{Z}$, be the roots of the quadratic equation
 $x(x + 2) + (x + 1)(x + 3) + (x + 2)(x + 4) + \dots + (x + n + 1)(x + n - 1) = 4n$ for some $n \in \mathbb{N}$. Then $n + \alpha$ is equal to :

- A) 0
- B) 1
- C) 2
- D) 3

{Solution: <https://www.youtube.com/watch?v=AVSUet9a0>}